

N+ Network security

Theme: Startup Office IT Setup

Section A: Scenario-Based Concept Application

1. Explain the selection process for CPU, RAM, and storage specifications to ensure hardware compatibility and prevent performance bottlenecks for a startup team performing concurrent development and video conferencing tasks.

Answer :- For a startup team, the computer should be able to run development software and video-conferencing applications at the same time.

1. CPU Selection

Select a modern multi-core CPU, such as Intel Core i5/i7 or AMD Ryzen 5/Ryzen 7.

A multi-core CPU is useful because it can handle multiple tasks at the same time.

2. RAM Selection

At least 16 GB RAM is suitable for development and video conferencing.

For heavy development, virtual machines, or multiple applications, 32 GB RAM is better.

More RAM helps applications run together without making the computer slow.

3. Storage Selection

Use an SSD, preferably an NVMe SSD.

Recommended capacity:

- 512 GB for normal development
- 1 TB for larger projects and files

An SSD provides faster booting, application loading and file access than an HDD.

4. Compatibility Checking

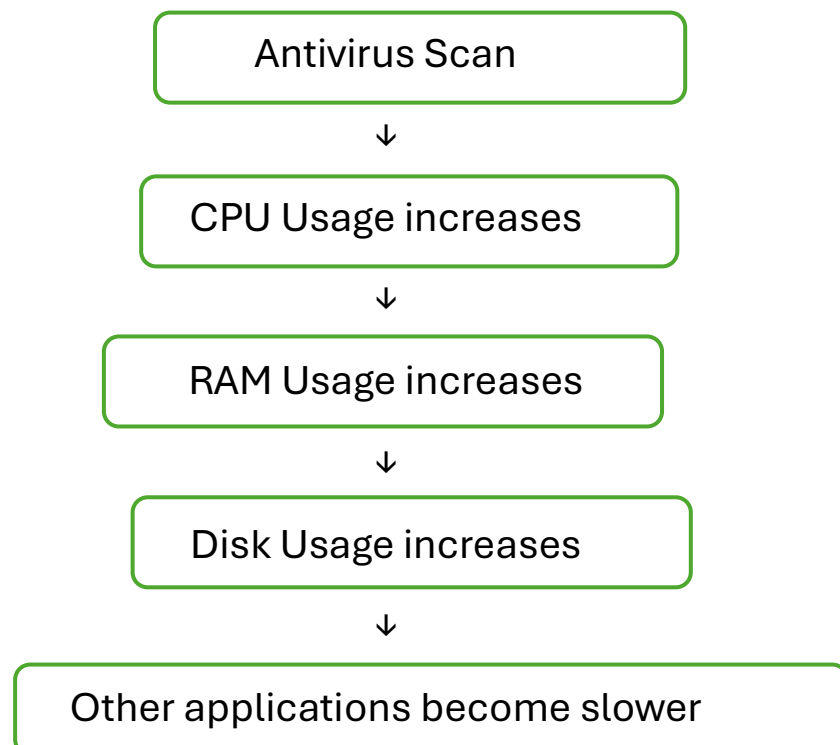
Before installing the components, check:

- CPU socket must match the motherboard.
- RAM type must be supported by the motherboard.
- RAM speed should be supported.
- SSD interface must be compatible.
- PSU must provide sufficient power.

Q2. Explain how system resource allocation and hardware bottlenecks contribute to performance degradation after the deployment of resource-intensive background applications like antivirus software.

Answer :-

Antivirus software normally runs in the background and uses system resources such as CPU, RAM and disk.



If the computer has insufficient RAM or a slow CPU/storage device, the performance may decrease further.

How to troubleshoot:

1. Open **Task Manager**.
2. Click **Processes**.
3. Check CPU usage.
4. Check Memory usage.
5. Check Disk usage.
6. Identify the application using excessive resources.
7. Disable unnecessary startup applications.
8. Schedule antivirus scans during less busy hours.
9. Upgrade RAM or other hardware if required.

3. Explain the systematic troubleshooting process for resolving peripheral hardware failures caused by driver incompatibility or operating system conflicts when a device functions inconsistently across different workstations.

Answer :-

Peripheral devices include printers, keyboards, mice, webcams, scanners and USB devices.

Step 1 – Check physical connection

(check :- USB cable, Power cable, USB port, Device power)

Step 2 – Try another port

Step 3 – Test another computer

Step 4 – Check Device Manager

Step 5 – Check driver

Step 6 – Reinstall driver

Step 7 – Check operating system compatibility

4. Explain the role of Uninterruptible Power Supply (UPS) systems and surge protection in preventing file system corruption and ensuring data integrity during unexpected power fluctuations.

Answer :-

A **UPS (Uninterruptible Power Supply)** provides temporary battery power when the main electricity supply fails.

A surge protector protects computer components from sudden increases in voltage.

Benefits:

- Protects computer hardware.
- Reduces the chance of file-system corruption.
- Prevents unexpected shutdowns.
- Helps protect important data.
- Provides time for safe shutdown.

Section B: Practical Application

Task 1 : Physically assemble a workstation and document the component installation sequence.

Answer :-

Components

- Motherboard ,CPU , RAM, CPU cooler, SSD/HDD, Power Supply (PSU), Cabinet, Monitor, Keyboard and mouse.

Simple Installation Steps :-

- Take the motherboard.
- Install the CPU.
- Install the CPU cooler.
- Insert the RAM.
- Install the SSD/HDD.
- Put the motherboard inside the cabinet.
- Install the PSU.
- Connect all power and data cables.
- Connect monitor, keyboard and mouse.
- Turn on the computer.
- Check whether the computer passes POST.

Task 2 : Troubleshoot and resolve a RAM mismatch error within the BIOS/UEFI settings.

Answer :-

Example :- RAM physically installed = 16 GB

BIOS showing = 8 GB

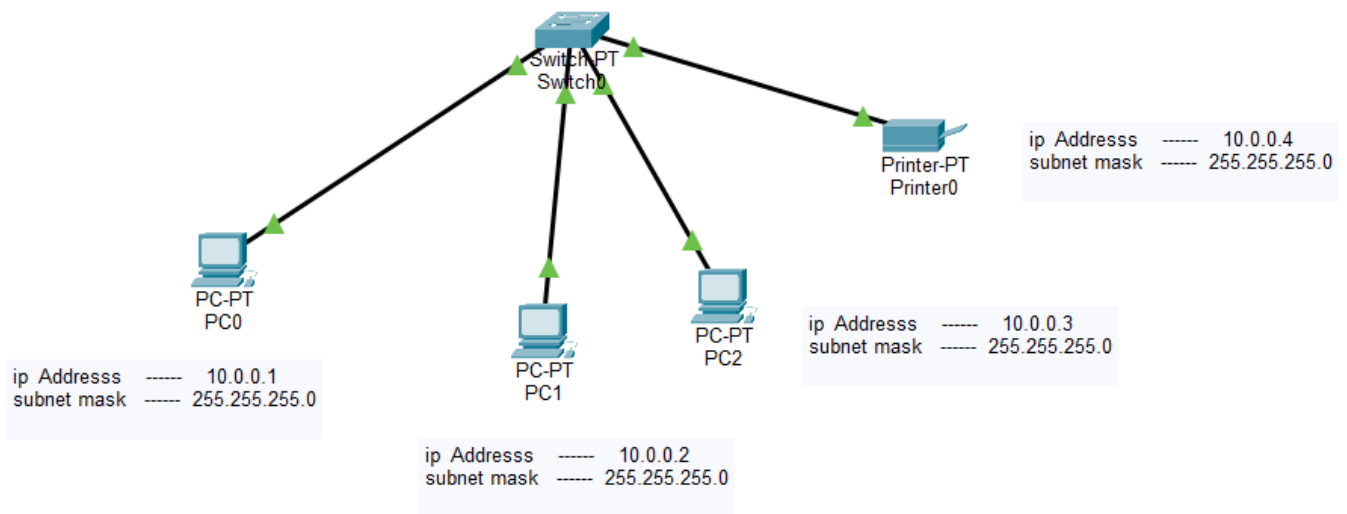
This means the BIOS is not detecting all the RAM.

Simple Steps

1. Restart the computer.
2. Press **F2 or Delete** to enter BIOS/UEFI.
3. Check the RAM/memory information.
4. Note the detected RAM.
5. Turn off the computer.
6. Remove and properly insert the RAM again.
7. Check whether the RAM is installed in the correct slot.
8. If there are two RAM sticks, test them one at a time.
9. Start the computer again.
10. Enter BIOS and check the RAM again.
11. If necessary, load **BIOS default settings**.
12. Save and restart.

Task 3 : Build a LAN topology in Packet Tracer for 5 systems and 1 printer using a Class C subnet.

Answer :-



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time=2ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time=1ms TTL=128

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 10.0.0.3

Pinging 10.0.0.3 with 32 bytes of data:

Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time=1ms TTL=128

Ping statistics for 10.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.0.0.4

Pinging 10.0.0.4 with 32 bytes of data:

Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time=2ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time=1ms TTL=128

Ping statistics for 10.0.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

Task 4 : Perform a clean OS installation with manual disk partitioning and startup optimization.

Answer :-

To install a fresh operating system, divide the hard disk into partitions, and make the computer start faster.

Steps

1. Start Installation

- Connect the OS installation USB.
- Start the computer from the USB.
- Select **Language, Time and Keyboard**.
- Click **Install Now**.

2. Choose Installation

- Select **Custom Installation**.
- This allows us to manage the disk manually.

3. Create Partitions

Example:

Hard Disk

C: Drive → OS/System

D: Drive → Personal Files

Recovery → Recovery Files

4. Install OS

- Select the **C: Drive**.
- Click **Next**.
- Wait for the installation to finish.

- The computer will restart.

5. Complete Setup

- Create a username and password.
- Select the required settings.
- Connect to the internet if needed.

Startup Optimization

After installing the OS:

- Open **Task Manager**.
- Click **Startup Apps**.
- Disable unnecessary programs.
- Keep important system and security programs enabled.

Benefits

- Fresh OS installation.
- Better disk organization.
- Faster startup.
- Removes unnecessary programs.
- Improves computer performance.

Section C: Integrated Case Study

Scenario: At Nexus Innovations, three rendering workstations pass morning POST checks but restart randomly during heavy afternoon workloads. The units are stored in poorly ventilated desk cabinets.

Questions:

1. Identify the three hardware components most likely to cause random reboots under load.
2. Explain why the systems pass POST in the morning but fail in the afternoon.
3. How would you use a stress-test tool to confirm a thermal or power issue?
4. Propose three specific steps to remediate the frequent system restarts.
5. Provide a brief technical justification for why a PSU upgrade might (or might not) be necessary.